

Maruf Ahmed Bhuiyan

+8801792514074 | dr.marufahmed92@gmail.com | ORCID: 0000-0003-4783-8735 | Portfolio

Career Objective

As a medical professional bridging virology, biotechnology, and public health, I specialize in infectious disease surveillance and molecular diagnostics. Leveraging practical clinical and high-containment laboratory experience, I aim to advance biomedical research and evidence-based public health solutions through data-driven genomic analysis and cross-sector collaboration.

Education

North South University

Dhaka, Bangladesh

Master of Public Health (MPH)

Jan 2026 - Ongoing

- CGPA: 4 on a scale of 4 (2nd semester)

Courses: Foundations in Epidemiology, Foundations in Biostatistics, Environmental Health, Public Health Information, Education & Communication.

Bangabandhu Sheikh Mujib Medical University

Dhaka, Bangladesh

Doctor of Medicine (MD) in Virology

Mar 2019 – Mar 2023

- 80% marks in thesis (Overall 75.62%)
- **Thesis Title:** Single nucleotide polymorphisms of VDR Apal, PNPLA3 I148M, TLR4 D299G in Hepatocellular carcinoma patients of Bangladesh
- **Courses:** General and Molecular Virology, Applied and Diagnostic Virology, Medical Immunology, Clinical Pathology, Medical Education, Systemic Virology, Clinical Virology and Viral Infectious Diseases, Bacteriology, Parasitology, Mycology, Biostatistics, Research Methodology

North South University

Dhaka, Bangladesh

Master of Science (MS) in Biotechnology

Sep 2019 – Sep 2022

- CGPA: 3.97 on a scale of 4 (Passed with distinction)
- **Courses:** Advanced Biochemistry and Molecular Biology, Advanced Cell Biology, Principles of Immunology, Bioinformatics, Biostatistics, Pharmaceutical Biotechnology, Virology, Industrial Microbiology, Quality Control Compliance, Genomic Sequencing and Analysis

Sylhet MAG Osmani Medical College

Sylhet, Bangladesh

Bachelor of Medicine, Bachelor of Surgery (MBBS)

Jan 2009 – July 2015

- **Courses:** Anatomy, Physiology, Biochemistry, Forensic Medicine, Community Medicine, Microbiology, Pharmacology, Pathology, Surgery, Medicine, Obstetrics and Gynaecology

Professional Experience

Characterizing the epidemiological diversity of Nipah strains, IEDCR

1 May 2025 – Current

Laboratory Research Officer

Mohakhali, Dhaka

- Performing routine laboratory test and molecular analysis such as molecular detection, and serology techniques such as PCR, ELISA, Sequencing etc. in BSL-2+ laboratory according to SOP.
- Contribution to writing reports and scientific manuscripts, conducting training and workshop.

- Organizing the laboratory, developing SOPs, supervising and monitoring the laboratory technologists, and providing support to maintain Biosafety and Biosecurity in the laboratory.

Head of the Department

3 July 2024 – 30 April 2025

Department of Microbiology, Shahabuddin Medical College

Dhaka, Bangladesh

- Develop and update academic curriculum, mentor students, and participate in examinations and academic assessments.
- Supervision of laboratory tests related to infectious diseases and reporting.
- Conducting examinations to evaluate students.

Assistant Professor

2 May 2023 – 30 April 2025

Department of Microbiology, Shahabuddin Medical College

Dhaka, Bangladesh

- Deliver lectures and practical training in microbiology, including bacteriology, virology, and immunology for medical students.
- Teaching and supervising undergraduate medical students about Microbiology.

Resident (MD)

March 2019 – Jan 2023

Department of Virology, Bangabandhu Sheikh Mujib Medical University (BSMMU)

Dhaka, Bangladesh

- Supervision of diagnostic testing of Hepatitis serological panel, HIV serological panel, molecular testing for HBV, HCV, HIV, CMV and HPV and various other viral diseases.
- Result analysis and reporting.

Medical Officer

Feb 2017 – Sep 2018

Neonatal Intensive Care Unit (NICU), Padma General Hospital

Dhaka, Bangladesh

- Monitoring & providing treatment to neonates.
- Counselling of parents regarding the status of their neonates.
- Leading a healthcare team of 6 including nurses and medical technologist to provide the best care to newborns.

Intern Doctor

Oct 2015 – Oct 2016

Sylhet MAG Osmani Medical College

Sylhet, Bangladesh

- Attending outdoor and emergency patients.
- Admission of patients following standard protocol.
- Doing necessary instrumentation for patient management e.g., Urinary catheter insertion, IV catheter insertion, Nasogastric tube insertion, Blood transfusion, wound dressing etc.

Training, Workshops & Internships

Writing for Public Health Bulletins Workshop

Virtual

CDC Foundation

9 March – 3 April, 2026

- Synthesizing complex epidemiological data into actionable insights.
- Translating complex epidemiological findings and public health data into clear, evidence-based narratives for diverse audiences.
- Applying best practices in scientific writing to support evidence-based advocacy.
- Structuring surveillance articles and formal public health bulletins for maximum impact and policy influence.

Certificate Course on Climate Change and Public Health

University of Dhaka

Dhaka

January – April, 2026

- Trained in climate change science, climate-sensitive diseases, and climate-resilient public health systems.
- Gained practical experience in climate-health surveillance, health impact assessment, and adaptation planning.
- Developed skills in climate-health research, project proposal development, policy analysis, and scientific communication.

Bioinformatics Internship Programme

HackBio

Virtual

1 Aug – 5 Sep, 2020

- Bash scripting, Phylogenetic analysis, RNA-seq analysis

Introduction to Programming in R for Biologist

Center for Bioinformatics Learning Advancement and Systematic Training (CBLAST), University of Dhaka

Dhaka, Bangladesh

26 Jun – 10 Jul, 2020

- R basic syntax, Data structures in R.
- Data exploration and wrangling, Visualization of data using ggplot2.

Basic Course on Structural Biology

Bangladeshi Structural Biologist

Dhaka, Bangladesh

26 Jun – 7 Jul, 2020

- Protein structure, Expression and protein purification, Protein crystallization and X-ray crystallography, Analysis of X-ray diffraction data.
- Cryo-electron microscopy, Small angle x-ray scattering (SAXS), NMR, Electron Paramagnetic Resonance (EPR), Model building and validation.

Cheminformatics: Drug Target Discovery and Computer-aided Drug Design

CBLAST, University of Dhaka

Dhaka, Bangladesh

01 April – 10 August, 2020

- Drug discovery and development, Pre-clinical research, Pharmacokinetics, Pharmacodynamics, Ligand preparation, Chemical compound database.
- Protein structure database, Active site prediction, Protein structure prediction, Homology modeling, QSAR, ADMET analysis, Virtual screening.

Training on Immunoinformatics and Vaccinomics

BioSol Center

Dhaka, Bangladesh

6 – 7 March, 2020

- Immuno-informatics analysis, Peptide based vaccine design, Homology modeling, Vaccine evaluation.
- Molecular docking, Molecular dynamics simulation, Disulfide engineering, In-silico cloning, Immune simulation.

Training on "CRISPR: Genome Engineering Technology"

Department of Pharmacy, Brac University in partnership with Merck KGaA, Darmstadt, Germany

Dhaka, Bangladesh

21 – 22 December, 2019

- Introduction to genome editing, Gene modification: ZFN & TALEN, Targeted integration using CRISPR.
- Bioinformatics for CRISPR, CRISPR workflow: Nuclease optimization & clonal isolation, Case studies in cell line engineering.

Certificate Course on Research Methodology

Dhaka, Bangladesh

Centre for Medical Research & Development (CMRD)

14 Sep – 16 Nov, 2019

- Fundamentals of research, Study designs, Sampling techniques.
- Descriptive statistics, Inferential statistics, Thesis writing.

Training on Animal Cell Culture Techniques for Advanced Research

Dhaka, Bangladesh

Centre for Advanced Research in Sciences (CARS), University of Dhaka

25 – 29 Aug, 2019

- Concept of cell culture, Cell type and characteristics, Safety Aspects of cell culture, Procedures of cell culture technique.
- Cell passage, Cell counting, Cell seeding, Cell preservation, Cell maintenance, Cell thawing.
- Cytotoxicity testing using N4X4 cell line, Quality control considerations, Cryopreservation and Storage of cells.

Advanced Training Workshop on Molecular Techniques

Dhaka, Bangladesh

Institute for Developing Science and Health Initiatives (ideSHi)

29 – 31 July, 2019

- DNA extraction for human and clinical bacterial isolates, DNA quantification.
- PCR: Conventional PCR, real-time PCR, High resolution melt curve analysis (HRM), Reverse Transcriptase PCR (RT-qPCR).
- Agarose gel electrophoresis.

Honors & Awards

2023	Bangabandhu Sheikh Mujib Medical University, University Thesis Grant (~ \$1000) for MD (Virology) thesis.
2026	Full Funded Scholarship (~ \$400) for Certificate Course in Climate Change and Public Health by DSCR (DU) & CCHPU (MoHFW) in collaboration with FCDO and UNFPA
2026	Scholarship (~ \$670) for Master of Public Health (MPH) at North South University

Skills

Wet Lab	Nucleic acid extraction, PCR (cPCR, qPCR, RT-qPCR), Electrophoresis, Cell culture
Programming	R, Shell (Bash/Zsh), Python
Data Analysis	Stata, Jamovi, R
Scientific Illustration	Microsoft PowerPoint, Adobe Illustrator, Inkscape, QGIS
Soft Skills	Engaging Presentation, Problem-solving, Report writing, Teaching, Training.
Miscellaneous	Microsoft Office, Git/GitHub, AI-native

Publications

PUBLISHED

Association of climate variability with hepatitis A and E infections in Dhaka (2016–2023)

SM Rashed Ul Islam, Afsana Anwar Miti, Ayesha Alam Mita, Ridwana Asma, **Maruf Ahmed Bhuiyan**, Saif Ullah Munshi, Afzalun Nessa. *Bangabandhu Sheikh Mujib Medical University Journal* 18 (3). e83258-e83258 (Sep 2025)

Nipah virus in Bangladesh - A 25-year Narrative Review of Epidemiology, Transmission and Control strategies

Reefat Fateema, **Maruf Ahmed Bhuiyan**. *Shahabuddin Medical College Journal* 10.2 (July 2025) pp. 24- 28.

Re- emergence of Mpox virus- A short review

Maruf Ahmed Bhuiyan, Reefat Fateema. *Shahabuddin Medical College Journal* 9.2 (July 2024) pp.31-34.

Use of whole blood and dried blood spot for detection of HIV-1 nucleic acids using reverse transcription loop-mediated isothermal amplification

Mohammad Jahidur Rahman Khan, **Maruf Ahmed Bhuiyan**, Shahina Tabassum, Saif Ullah Munshi. *Journal of Virological Methods* 312 (Feb. 2023) p. 114642.

Choice of assemblers has a critical impact on de novo assembly of SARS-CoV-2 genome and characterizing variants

Rashedul Islam, Rajan Saha Raju, Nazia Tasnim, Istiak Hossain Shihab, **Maruf Ahmed Bhuiyan**, Yusha Araf, Tofazzal Islam. *Briefings in Bioinformatics* 22.5 (Oct. 2021) bbab102.

Discovery of potential immune epitopes and peptide vaccine design – a prophylactic strategy against Rift Valley fever virus

Maruf Ahmed Bhuiyan, Syeda Tasnim Quayum, Foysal Ahammad, Rahat Alam, Abdus Samad, Zulkar Nain. *F1000Research* 9 (Aug. 2020) p. 999.

PRE-PRINTS & MANUSCRIPTS IN PREPARATION

Discovery of Potential Compounds Against Nipah Virus: A Molecular Docking and Dynamics Simulation Approaches

Maruf Ahmed Bhuiyan, Nawshin Atia Keya, Farha Susan Mou, Raihan Rahman Imon, Rahat Alam, Abdus Samad, Foysal Ahammad. *Pre-print, Awaiting journal submission* (Jan. 2022).

Single nucleotide polymorphisms of VDR Apal, PNPLA3 I148M, TLR4 D299G in Hepatocellular carcinoma patients of Bangladesh

Maruf Ahmed Bhuiyan, Saif Ullah Munshi. *Thesis, Awaiting publication* (Mar. 2023).

Professional Membership & Registrations

Bangladesh Medical and Dental Council (BMDC)

Registered Physician (A-79860)

Society for Medical Virologists, Bangladesh (SMV,B)

General Member

American Society for Microbiology (ASM)

Global Outreach - Contributing Member

American Society for Virology (ASV)

Associate Member

Languages

Bangla	Native proficiency
English	Professional proficiency, IELTS 8 (L-8.5, R-8.5, W-7, S-7)
Hindi	Conversational proficiency
Spanish	Basic proficiency (CEFR A1)

References

Prof. Dr. Saif Ullah Munshi	Department of Virology, Bangabandhu Sheikh Mujib Medical University, Shahbag, Dhaka, Bangladesh. Contact: 01711-376343 Email: saifmunshi@yahoo.com
Prof. Dr. Mahmud Hossain	Department of Biochemistry and Molecular Biology, Dhaka University; Guest Faculty, Department of Biochemistry and Microbiology, North South University, Dhaka, Bangladesh. Contact: 01712-659151 Email: mahmud.hossain02@northsouth.edu
Prof. Dr. Khademul Azad	Principal (In-Charge) & Head of the Department of Biochemistry, Shahabuddin Medical College, Gulshan-2, Dhaka, Bangladesh. Contact: 8801924-798849 Email: khademulazad@gmail.com